

RICH LAKE AGDM, AB, Canada

WMO: 713490

Lat: 54.5012N

Lon: 111.6914W

Elev: 616

StdP: 94.14

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.8	-30.3	-37.3	0.1	-33.6	-33.5	0.2	-30.0	9.5	-4.2	8.7	-5.2	1.8	280	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	27.7	17.8	25.8	16.8	24.1	16.0	19.2	25.7	17.9	24.0	16.8	22.4	3.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.9	22.6	15.5	11.9	20.6	14.5	11.1	19.3	57.2	25.8	52.9	24.1	49.5	22.5	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.6	-38.4	30.5	3.5	1.9	-40.9	31.9	-43.0	33.0	-44.9	34.1	-47.5	35.4		
(5)			-38.5	21.5	3.5	1.0	-41.0	22.2	-43.1	22.8	-45.0	23.3	-47.6	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-13.0	-12.2	-5.9	2.5	9.6	14.0	16.5	14.7	9.7	2.9	-6.2	-13.0		
	DBStd	12.40	8.99	7.95	7.53	5.44	4.53	3.13	2.84	3.20	4.39	4.77	6.97	8.30		
	HDD10.0	3607	714	621	492	231	63	6	0	3	57	225	484	712		
	HDD18.3	6092	972	854	751	476	272	134	71	119	259	479	734	970		
	CDD10.0	582	0	0	0	4	50	124	202	148	48	4	0	0		
	CDD18.3	24	0	0	0	0	1	3	14	5	1	0	0	0		
	CDH23.3	534	0	0	0	1	37	96	234	125	41	2	0	0		
	CDH26.7	102	0	0	0	0	5	18	54	17	8	0	0	0		
(14) Wind	WSAvg	3.0	2.9	2.9	3.3	3.5	3.4	2.9	2.7	2.7	2.9	3.2	3.1	2.8		
(15) Precipitation	PrecAvg	437	19	14	18	31	43	76	83	62	36	22	20	19		
	PrecMax	526	50	32	35	86	81	162	183	138	92	56	56	36		
	PrecMin	278	0	1	5	2	4	22	32	22	3	5	2	1		
	PrecStd	67	11	7	9	17	21	33	28	26	22	11	14	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	7.0	11.3	21.7	26.9	28.7	30.8	28.8	27.6	21.8	11.0	5.8		
		MCWB	2.8	2.5	5.5	10.6	14.4	17.3	19.8	18.4	16.4	12.5	5.5	2.4		
	2%	DB	4.4	4.0	8.3	17.9	23.8	26.0	28.1	26.4	23.8	15.9	7.1	3.5		
		MCWB	1.4	0.9	3.5	8.5	12.9	15.8	19.2	17.6	14.7	9.4	3.2	0.8		
	5%	DB	2.2	1.6	6.0	14.5	21.5	23.6	25.9	24.6	20.8	12.7	4.7	1.1		
		MCWB	0.1	-0.8	2.1	6.7	11.5	14.7	17.7	17.0	13.4	7.2	1.9	-1.0		
	10%	DB	-0.4	-0.7	3.7	11.6	19.2	21.7	23.9	22.6	17.8	10.3	2.4	-1.5		
		MCWB	-1.9	-2.4	0.7	5.2	10.3	13.8	16.9	15.9	11.9	5.9	0.4	-2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.9	5.8	11.3	15.6	18.6	21.3	20.4	17.0	12.7	6.1	2.7		
		MCDB	6.1	6.0	10.8	21.0	24.1	26.9	27.8	26.2	25.7	20.2	10.5	5.5		
	2%	WB	1.6	1.0	3.7	9.0	13.5	17.0	19.8	18.5	15.3	9.7	3.9	1.0		
		MCDB	4.0	3.7	7.7	17.1	21.6	23.8	26.3	24.6	22.5	14.7	6.6	3.4		
	5%	WB	0.2	-0.6	2.3	7.0	12.3	15.9	18.6	17.3	13.8	7.7	2.1	-0.9		
		MCDB	2.3	1.4	5.6	13.9	19.9	21.9	24.7	23.2	20.0	12.0	4.4	0.9		
	10%	WB	-1.9	-2.4	1.0	5.5	11.1	14.7	17.5	16.3	12.5	6.3	0.4	-2.8		
		MCDB	-0.3	-0.6	3.4	10.9	17.8	20.1	22.8	21.8	17.1	9.7	2.1	-1.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.2	11.0	11.6	13.4	12.3	12.2	12.5	12.1	10.0	8.4	9.4		
		MCDBR	10.8	13.2	12.9	17.1	17.5	16.4	15.9	16.3	17.9	15.0	10.7	11.8		
		MCWBR	8.3	10.6	8.7	9.5	8.4	7.9	7.9	8.4	9.4	9.0	7.5	9.7		
		MCDBR	10.8	13.0	12.3	16.1	15.6	14.3	14.7	15.0	16.7	14.3	10.3	11.6		
	5% WB	MCWBR	8.5	10.6	8.5	9.3	8.0	7.5	7.8	8.1	9.0	8.9	7.6	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.260	0.288	0.325	0.342	0.356	0.358	0.363	0.310	0.280	0.252	0.234		
	taud		2.271	2.315	2.356	2.394	2.402	2.373	2.423	2.354	2.517	2.574	2.449	2.281		
	Ebn at Noon		716	846	897	904	903	890	882	852	868	827	732	663		
	Edh at Noon		56	76	92	103	108	113	106	107	79	59	47	44		
(44) All-Sky Solar Radiation	RadAvg		0.85	1.88	3.41	4.65	5.67	5.83	5.91	4.81	3.37	1.87	0.87	0.59		
	RadStd		0.09	0.15	0.24	0.38	0.48	0.40	0.32	0.38	0.36	0.30	0.11	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		-0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page